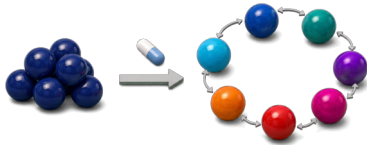


Drug-induced cell-state cycling, detected as a topological loop and measured by persistent homology

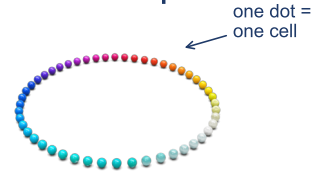
(a) drug pressure induces cycling → (b) scRNA-seq sees it as a loop → (c) persistent homology measures the loop's lifespan

a · biology



Drug pressure makes cancer cells cycle between states

b · scRNA-seq



scRNA-seq sees the cycle as a closed loop



quantify with persistent homology

c · how we measure the loop

How long the loop survives
(max H_1 persistence)

